

Financial Analytics Lab

Assignment-1

Group 11

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Financial ratio analysis is crucial for assessing a company's financial health and performance. Ratios provide insights into a firm's *efficiency, profitability, liquidity, and solvency*. Investors, creditors, and management use these ratios to make informed decisions, identify trends, and evaluate risk. Key ratios include liquidity ratios (e.g., current ratio), profitability ratios (e.g., return on equity), and leverage ratios (e.g., debt-to-equity). Overall, financial ratio analysis is a powerful tool for informed decision-making in finance and business. Here we analyse some of the crucial ratios for all the 5 firms across all years -

Link to our Excel Worksheet for all calculations: [Team 11 - Main Sheet](#)

Question 1

Current Ratio :

Current Ratio	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.643	0.669	0.189	0.855	0.916
Mar '22	0.330	0.637	0.112	0.724	0.899
Mar '21	0.248	0.679	0.045	0.485	0.799
Mar '20	0.241	0.498	0.212	0.621	0.820
Mar '19	0.318	0.585	0.060	0.455	0.772

The current ratio is a liquidity ratio that measures a company's ability to cover its short-term liabilities with its short-term assets. A ratio above 1 indicates that a company has more current assets than current liabilities, suggesting good liquidity.

Observations:

- IOCL and NTPC: Both IOCL and NTPC have high current ratios (0.916 and 0.855, respectively), indicating a strong ability to cover short-term obligations. They have a comfortable margin of current assets to meet their current liabilities.
- BPCL and ONGC: BPCL and ONGC also have current ratios below 1, but slightly better than CIL. This suggests that they may face challenges in covering their short-term obligations with their current assets. However, they are in a better position than CIL.
- CIL: Coal India Limited (CIL) has the lowest current ratio at 0.189. This may imply potential difficulty in meeting short-term liabilities with available current assets. It's worth investigating the reasons behind this low ratio, such as inventory management or outstanding liabilities.

Quick Ratio:

Quick Ratio	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.510	0.126	0.186	0.578	0.115
Mar '22	0.200	0.144	0.110	0.524	0.140
Mar '21	0.121	0.243	0.045	0.308	0.129
Mar '20	0.097	0.102	0.210	0.388	0.143
Mar '19	0.171	0.140	0.054	0.259	0.138

The quick ratio, also known as the acid-test ratio, is a more stringent measure of liquidity than the current ratio. It excludes inventory from current assets since inventory may not be as easily converted to cash. A quick ratio above 1 indicates a company's ability to cover short-term liabilities with its most liquid assets.

Observations:

- NTPC and ONGC: NTPC has the highest quick ratio at 0.578, followed by ONGC at 0.510. Both companies have a relatively better ability to cover short-term obligations without relying heavily on inventory.
- Such a large change in the Current and Quick Ratio of BPCL and IOCL depicts that there are considerable amounts of inventories on the balance sheet which are not very strictly liquid.
- CIL and BPCL: Both CIL and BPCL have quick ratios below 0.2, indicating potential challenges in meeting short-term obligations without relying on the sale of inventory. BPCL's quick ratio is particularly low at 0.126, raising concerns about its liquidity position.
- IOCL: Indian Oil Corporation Limited (IOCL) has the lowest quick ratio at 0.115, suggesting a potential difficulty in covering short-term liabilities with highly liquid assets. Further investigation is needed to understand the specific factors contributing to this low ratio.

Return on Assets:

Return on Assests	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.153	0.023	0.894	0.055	0.032
Mar '22	0.174	0.117	0.677	0.055	0.108
Mar '21	0.053	0.259	0.455	0.049	0.106
Mar '20	0.062	0.040	0.732	0.039	0.007
Mar '19	0.119	0.112	0.748	0.020	0.088

Return on Assets (ROA) is a financial ratio that measures a company's ability to generate profit from its assets. It is calculated by dividing net income by average total assets. A higher ROA indicates better efficiency in utilising assets to generate earnings.

Observations:

- CIL (Coal India Limited): CIL stands out with the highest ROA at 0.894, indicating that it is highly effective in generating profit relative to its total assets. This could be attributed to efficient management, cost control, or successful utilisation of resources.
- ONGC (Oil and Natural Gas Corporation): ONGC has a moderate ROA of 0.153, suggesting a reasonable ability to generate profit from its assets. Further investigation into operational efficiency and industry benchmarks can provide more context.
- NTPC (National Thermal Power Corporation): NTPC has an ROA of 0.055, indicating a relatively lower efficiency in generating profit compared to CIL. Exploring the reasons behind this and comparing with industry peers can provide insights into potential areas for improvement.
- BPCL (Bharat Petroleum Corporation Limited) and IOCL (Indian Oil Corporation Limited): Both BPCL and IOCL have lower ROAs at 0.023 and 0.032, respectively. This suggests they may face challenges in generating profits relative to their total assets. Analysing factors such as cost management, revenue streams, and industry dynamics can help identify improvement opportunities.

Return on Equity:

Return on Equity	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.037	0.157	0.895	0.031	0.062
Mar '22	0.169	0.183	0.677	0.031	0.200
Mar '21	0.434	0.056	0.455	0.000	0.214
Mar '20	0.077	0.068	0.732	0.004	0.013
Mar '19	0.194	0.132	0.748	0.092	0.155

Return on Equity (ROE) is a financial ratio that measures a company's ability to generate profit from shareholders' equity. It is calculated by dividing net income by average shareholders' equity. A higher ROE indicates better profitability in relation to shareholders' investment.

Observations:

- CIL (Coal India Limited): CIL stands out with the highest ROE at 0.895, indicating that it is highly effective in generating profit from shareholders'

equity. This suggests strong financial performance and efficient utilisation of equity.

- BPCL (Bharat Petroleum Corporation Limited): BPCL also has a relatively high ROE at 0.157, reflecting good profitability in relation to shareholders' equity. This suggests effective management and the ability to deliver returns to shareholders.
- IOCL (Indian Oil Corporation Limited): IOCL follows with an ROE of 0.062, indicating reasonable profitability in relation to shareholders' equity. While not as high as CIL or BPCL, it still suggests effective use of equity.
- ONGC (Oil and Natural Gas Corporation) and NTPC (National Thermal Power Corporation): ONGC and NTPC have lower ROEs at 0.037 and 0.031, respectively. This indicates that they may be less efficient in generating profit from shareholders' equity, and further analysis is needed to understand the contributing factors.

Investor Perspective: High ROE values, particularly for CIL and BPCL, may be seen positively by investors, signalling strong returns on their equity investments.

Earnings Per Share:

Earning per share	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	30.860	8.780	24.020	143.230	5.980
Mar '22	32.040	41.270	18.180	132.060	26.340
Mar '21	8.940	90.980	12.400	122.710	23.780
Mar '20	10.690	13.640	18.310	114.780	1.430
Mar '19	21.240	36.260	16.990	108.550	18.400

Earnings Per Share (EPS) is a financial metric that represents the portion of a company's profit allocated to each outstanding share of common stock. It is calculated by dividing the net income by the number of outstanding shares. A higher EPS generally indicates higher profitability on a per-share basis.

Observations:

- NTPC (National Thermal Power Corporation): NTPC stands out with the highest EPS at 143.23, indicating substantial earnings per share. This suggests strong profitability on a per-share basis.

- ONGC (Oil and Natural Gas Corporation) and CIL (Coal India Limited): ONGC and CIL have EPS values of 30.860 and 24.020, respectively, indicating healthy earnings on a per-share basis. These companies seem to be generating significant profits relative to the number of outstanding shares.
- BPCL (Bharat Petroleum Corporation Limited) and IOCL (Indian Oil Corporation Limited): BPCL and IOCL have lower EPS values at 8.78 and 5.98, respectively. This suggests a lower level of earnings per share compared to the other companies, indicating potential variations in profitability.
- From the data we observed that the Earning per Share has increased for some of the companies and decreased for others over the year with NTPC boasting the highest increase from 108.550 to 143.20.

Price to Earnings Ratio:

Price-Earning Ratio	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	39.214	5.006	8.909	9.870	12.918
Mar '22	8.699	5.115	10.066	8.303	3.014
Mar '21	4.693	11.521	10.565	7.535	2.579
Mar '20	23.167	6.268	7.537	8.219	38.063
Mar '19	10.976	7.556	14.008	11.448	5.909

The Price-to-Earnings (P/E) ratio is a financial metric that compares a company's current stock price to its earnings per share (EPS). It is a key valuation ratio and provides insights into how the market values a company's earnings.

Observations:

- ONGC (Oil and Natural Gas Corporation): ONGC has the highest P/E ratio at 39.214. This implies that investors are willing to pay a higher multiple of earnings for each share of ONGC. It could indicate high expectations for future growth or positive market sentiment.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has the lowest P/E ratio at 5.006. A lower P/E ratio may suggest that the market values BPCL's earnings less compared to other companies in the list. This could be due to various factors such as perceived risks, lower growth expectations, or market conditions.

- CIL (Coal India Limited), NTPC (National Thermal Power Corporation), and IOCL (Indian Oil Corporation Limited): These companies fall in the middle range of P/E ratios, with CIL having a higher ratio compared to NTPC and IOCL. Investors seem to value CIL's earnings more relative to its stock price.

Profit Margin:

Profit Margin	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.004	0.269	7.589	0.105	0.010
Mar '22	0.024	0.365	9.896	0.139	0.040
Mar '21	0.082	0.165	11.933	0.139	0.058
Mar '20	0.009	0.140	13.348	0.104	0.003
Mar '19	0.024	0.244	11.206	0.130	0.032

Profit Margin is a financial metric that measures a company's profitability by expressing its net income as a percentage of total revenue. A higher profit margin indicates better efficiency in converting sales into profit.

Observations:

- CIL (Coal India Limited): CIL has an exceptionally high profit margin of 758.9%, indicating that it is very efficient in converting sales into profit. This could be due to cost controls, pricing strategies, or other factors that contribute to high profitability.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has a relatively high profit margin of 26.9%, indicating strong efficiency in converting revenue into profit. This suggests effective cost management and pricing strategies.
- NTPC (National Thermal Power Corporation): NTPC has a profit margin of 10.5%, which is a solid figure, suggesting a healthy conversion of sales into profit. It indicates reasonable efficiency in managing costs and generating profits.
- IOCL (Indian Oil Corporation Limited): IOCL has a profit margin of 1.0%, which is lower compared to BPCL and NTPC but still signifies a positive ability to convert sales into profit. Further analysis is needed to understand the factors contributing to this margin.
- ONGC (Oil and Natural Gas Corporation): ONGC has the lowest profit margin at 0.4%, indicating lower efficiency in converting sales into profit. It may be

worth exploring factors affecting profitability, such as operating costs or market conditions.

Total Asset Turnover:

Total Assets Turnover	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.568	5.854	0.118	0.526	3.296
Mar '22	0.476	4.843	0.068	0.399	2.663
Mar '21	0.318	3.159	0.038	0.356	1.828
Mar '20	0.444	4.202	0.055	0.378	2.438
Mar '19	0.488	4.649	0.067	0.372	2.750

Total Asset Turnover is a financial metric that measures a company's efficiency in generating revenue from its total assets. It is calculated by dividing the total revenue by the average total assets. A higher total asset turnover generally indicates better efficiency in utilising assets to generate sales.

Observations:

- BPCL (Bharat Petroleum Corporation Limited): BPCL has the highest total asset turnover at 5.854, suggesting exceptional efficiency in generating revenue from its total assets. This could be attributed to effective management and utilisation of assets.
- IOCL (Indian Oil Corporation Limited): IOCL also demonstrates a high total asset turnover of 3.296, indicating efficient utilisation of assets to generate sales. This suggests effective operational management.
- NTPC (National Thermal Power Corporation): NTPC has a moderate total asset turnover of 0.526, indicating reasonable efficiency in converting assets into revenue. Further analysis can help understand the factors influencing this ratio.
- ONGC (Oil and Natural Gas Corporation): ONGC has a total asset turnover of 0.568, which is moderate and suggests reasonable efficiency in generating revenue from its assets.
- CIL (Coal India Limited): CIL has the lowest total asset turnover at 0.118, indicating lower efficiency in generating revenue from its total assets. Further investigation is needed to understand the factors influencing this ratio.

Debt Ratio:

Debt Ratio	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.027	0.408	0.002	0.571	0.496
Mar '22	0.026	0.327	0.000	0.571	0.458
Mar '21	0.068	0.281	0.000	0.580	0.466
Mar '20	0.067	0.535	0.000	0.586	0.546
Mar '19	0.096	0.425	0.000	0.557	0.434

The Debt Ratio is a financial metric that measures the proportion of a company's assets financed by debt. It is calculated by dividing total debt by total assets. A lower debt ratio indicates lower financial leverage and potentially less financial risk.

Observations:

- CIL (Coal India Limited): CIL has an extremely low debt ratio of 0.002 (0.2%), indicating minimal reliance on debt to finance its assets. This suggests a conservative capital structure with a lower financial risk.
- ONGC (Oil and Natural Gas Corporation): ONGC also has a low debt ratio of 0.027 (2.7%), indicating a conservative approach to debt financing. This reflects a lower level of financial leverage.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has a relatively high debt ratio of 0.408 (40.8%), suggesting a significant portion of its assets are financed by debt. This may indicate higher financial leverage and increased financial risk compared to CIL and ONGC.
- IOCL (Indian Oil Corporation Limited): IOCL has a debt ratio of 0.496 (49.6%), indicating a substantial reliance on debt to finance its assets. This suggests a higher level of financial leverage and potential financial risk.
- NTPC (National Thermal Power Corporation): NTPC has the highest debt ratio at 0.571 (57.1%), indicating a significant portion of its assets is financed by debt. This suggests higher financial leverage and potentially higher financial risk compared to the other companies in the comparison.

Debt to Equity:

Debt to Equity	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.028	0.690	0.002	1.332	0.983
Mar '22	0.027	0.486	0.000	1.333	0.844
Mar '21	0.073	0.390	0.000	1.379	0.874
Mar '20	0.072	1.150	0.000	1.414	1.202
Mar '19	0.106	0.741	0.000	1.259	0.766

The Debt to Equity ratio is a financial metric that measures the proportion of a company's financing that comes from debt in relation to equity. It is calculated by dividing total debt by total equity. A lower ratio indicates less reliance on debt for financing, while a higher ratio suggests higher financial leverage.

Observations:

- CIL (Coal India Limited): CIL has an extremely low Debt to Equity ratio of 0.002 (0.2%), indicating minimal reliance on debt for financing. This suggests a conservative capital structure with a strong reliance on equity.
- ONGC (Oil and Natural Gas Corporation): ONGC also has a low Debt to Equity ratio of 0.028 (2.8%), reflecting a conservative approach to debt financing. The company relies more on equity to fund its operations.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has a relatively high Debt to Equity ratio of 0.690 (69.0%), indicating a significant reliance on debt for financing. This suggests a higher level of financial leverage compared to CIL and ONGC.
- IOCL (Indian Oil Corporation Limited): IOCL has a Debt to Equity ratio of 0.983 (98.3%), indicating a substantial reliance on debt. The company's capital structure involves a higher proportion of debt compared to equity.
- NTPC (National Thermal Power Corporation): NTPC has the highest Debt to Equity ratio at 1.332 (133.2%), suggesting a significant reliance on debt for financing. This indicates a higher financial leverage and potential financial risk compared to the other companies in the comparison.

Interest Coverage Ratio:

Interest Coverage Ratio	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	1.603	19.655	7862.203	3.345	2.389
Mar '22	7.392	18.379	7572.227	3.773	7.452
Mar '21	17.976	8.392	4831.025	3.089	10.610
Mar '20	2.139	8.112	2149.148	3.803	0.363
Mar '19	8.810	16.914	586.500	2.833	6.819

The Interest Coverage Ratio (ICR) is a financial metric that measures a company's ability to meet its interest obligations on outstanding debt. It is calculated by dividing a company's earnings before interest and taxes (EBIT) by its interest expenses. A higher ICR indicates a greater ability to cover interest payments and suggests lower financial risk.

Observations:

- CIL (Coal India Limited): CIL has an exceptionally high Interest Coverage Ratio of 7862.203, indicating an extremely strong ability to cover interest expenses. This suggests minimal financial risk related to debt servicing.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has a high Interest Coverage Ratio of 19.655, signifying a strong ability to cover interest payments. This indicates a robust financial position and lower financial risk.
- NTPC (National Thermal Power Corporation): NTPC has an Interest Coverage Ratio of 3.345, suggesting a moderate ability to cover interest expenses. While still indicating an ability to meet obligations, it is lower than BPCL.
- IOCL (Indian Oil Corporation Limited): IOCL has an Interest Coverage Ratio of 2.389, indicating a moderate ability to cover interest payments. It is lower than both BPCL and NTPC, suggesting comparatively higher financial risk.
- ONGC (Oil and Natural Gas Corporation): ONGC has the lowest Interest Coverage Ratio at 1.603, indicating a relatively lower ability to cover interest expenses. This raises concerns about the company's financial risk related to debt servicing.

Du-Pont Analysis:

DuPont Analysis	ONGC	BPCL	CIL	NTPC	IOCL
Mar '23	0.157	0.037	0.895	0.129	0.062
Mar '22	0.183	0.169	0.677	0.130	0.200
Mar '21	0.056	0.434	0.455	0.118	0.214
Mar '20	0.068	0.077	0.732	0.092	0.013
Mar '19	0.132	0.194	0.748	0.109	0.155

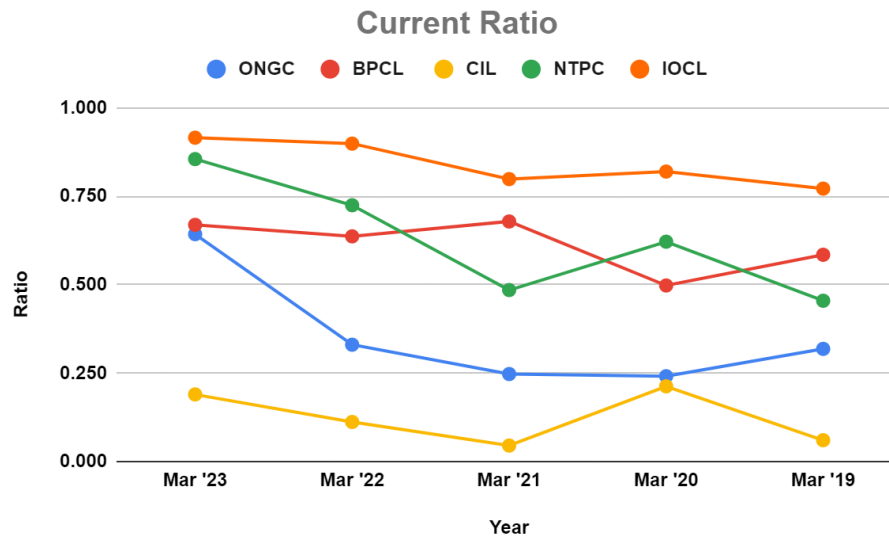
DuPont Analysis is a financial performance measurement tool that breaks down the Return on Equity (ROE) into three components: Net Profit Margin, Asset Turnover, and Equity Multiplier. The formula for DuPont Analysis is:

$$\text{ROE} = \text{Net Profit Margin} \times \text{Asset Turnover} \times \text{Equity Multiplier}$$

Observations:

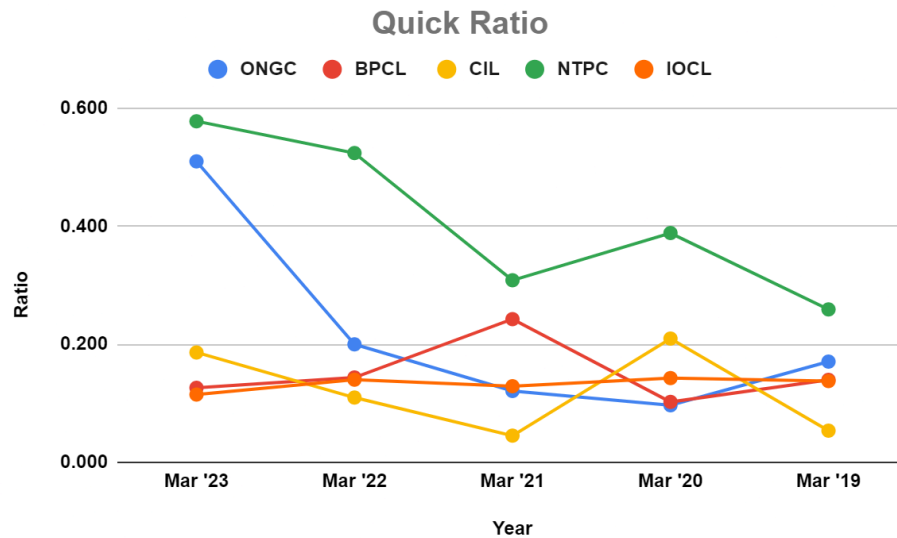
- CIL (Coal India Limited): CIL has the highest ROE at 0.895, indicating strong profitability and efficient utilisation of assets and equity. This high ROE could be attributed to a combination of a high Net Profit Margin, Asset Turnover, and/or Equity Multiplier.
- ONGC (Oil and Natural Gas Corporation): ONGC has an ROE of 0.157, suggesting a lower ROE compared to CIL. This could be due to factors such as a lower Net Profit Margin, Asset Turnover, or Equity Multiplier.
- NTPC (National Thermal Power Corporation): NTPC has an ROE of 0.129, indicating reasonable profitability and asset utilisation. The specific combination of Net Profit Margin, Asset Turnover, and Equity Multiplier contributes to this ROE.
- IOCL (Indian Oil Corporation Limited): IOCL has an ROE of 0.062, suggesting a lower ROE compared to CIL. This could be influenced by a combination of factors affecting Net Profit Margin, Asset Turnover, and Equity Multiplier.
- BPCL (Bharat Petroleum Corporation Limited): BPCL has the lowest ROE at 0.037, indicating comparatively lower profitability and asset utilisation. The specific factors affecting Net Profit Margin, Asset Turnover, and Equity Multiplier contribute to this lower ROE.

Question 2



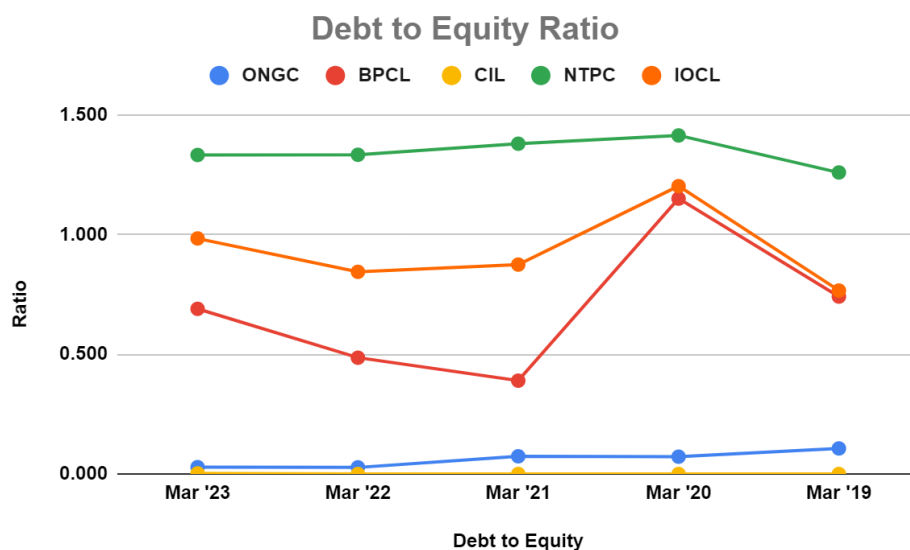
Inferences:

- **Liquidity Strength:** IOCL and NTPC appear to be in a strong liquidity position, indicating a good ability to meet short-term obligations.
- **Moderate Liquidity:** BPCL and ONGC have current ratios below 1, suggesting a need for careful management of short-term liquidity.
- **Concern for CIL:** CIL stands out with the lowest current ratio, raising concerns about its short-term liquidity. Further analysis is needed to understand the specific factors contributing to this low ratio.
- **From the data we observed that the current ratio has increased for all the companies over the year with NTPC boasting the highest increase from 0.455 to 0.855.**



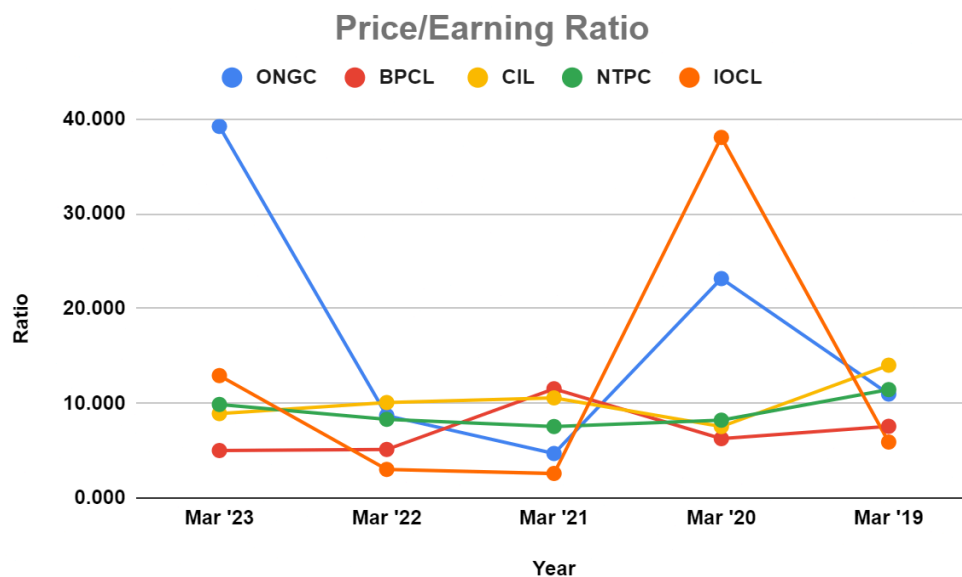
Inferences:

- **Liquidity Strength:** NTPC and ONGC appear to have better liquidity positions, with higher quick ratios compared to the other companies.
- **Concerns for BPCL and IOCL:** BPCL and IOCL show lower quick ratios, indicating potential challenges in covering short-term obligations without relying on inventory. It's essential to assess the reasons behind these low ratios.
- **From the data we observed that the quick ratio has increased for some companies and decreased for some companies over the year with NTPC boasting the highest increase from 0.171 to 0.591.**



Inferences:

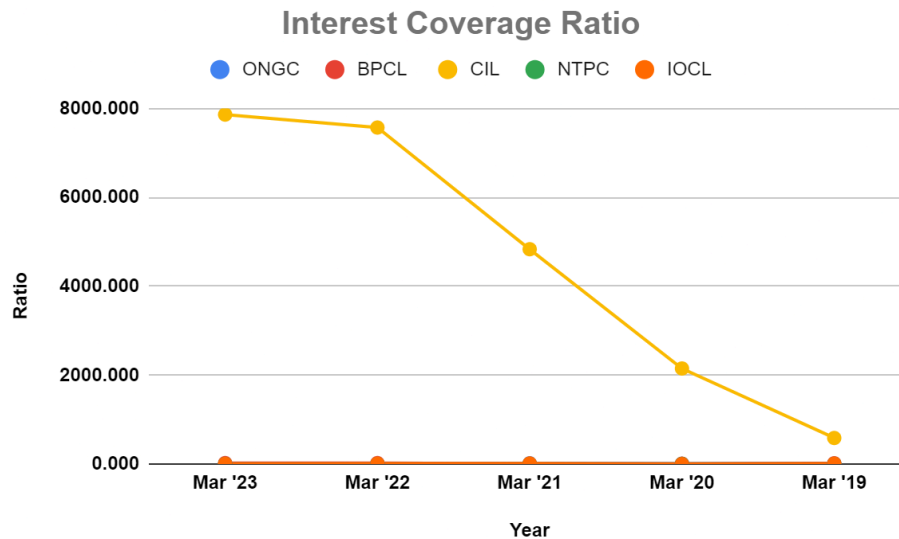
- CIL and ONGC's Conservative Financing: CIL and ONGC exhibit a conservative financing approach with very low Debt to Equity ratios, indicating a preference for equity financing and lower financial risk.
- BPCL, IOCL, and NTPC's Higher Leverage: BPCL, IOCL, and NTPC have higher Debt to Equity ratios, suggesting a greater reliance on debt for financing. This can offer tax advantages but also increases financial risk.
- From the data we observed that the Debt to Equity has increased for some of the companies and decreased for others over the year with IOCL boasting the highest increase.



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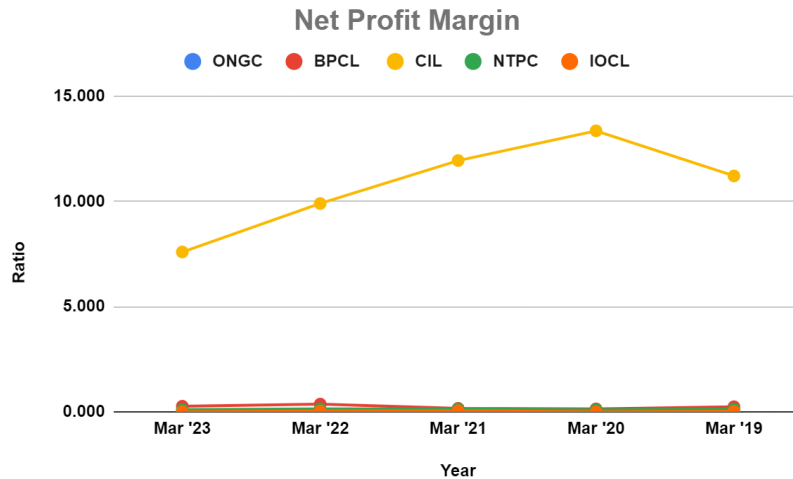
- Market Perception of Growth: A higher P/E ratio for ONGC may suggest that investors have high expectations for future growth or have a positive outlook on the company's prospects.
- BPCL's Lower Valuation: The lower P/E ratio for BPCL may indicate that the market values its earnings less compared to other companies. This could be influenced by factors such as perceived risks or lower growth expectations.
- CIL's Relatively Higher Valuation: CIL, with a P/E ratio of 8.909, is valued more highly compared to NTPC and IOCL. This suggests that investors may have more confidence in CIL's earnings potential.

- From the data we observed that the Price to Earning ratio has increased for some of the companies and decreased for others over the year with ONGC boasting the highest increase from 10.976 to 39.214.



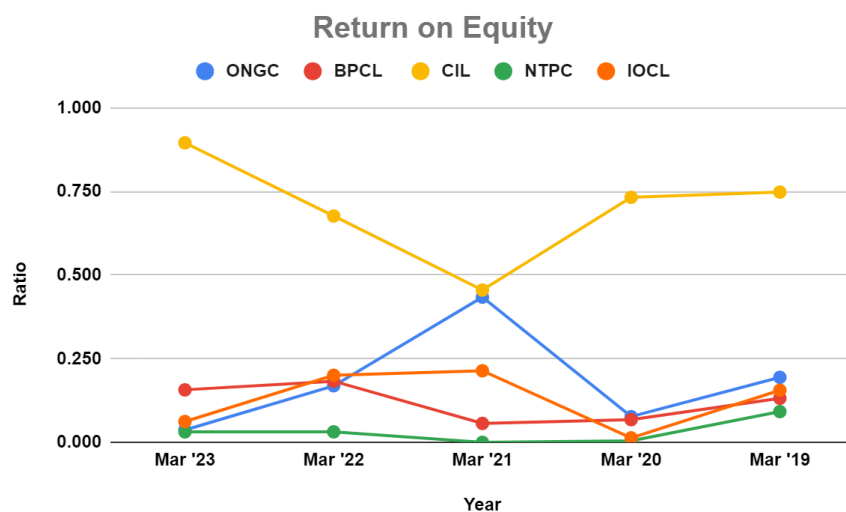
Inferences:

- CIL's Exceptional Financial Strength:** CIL stands out with an extraordinarily high Interest Coverage Ratio, demonstrating an extremely strong financial position and minimal financial risk related to interest payments.
- BPCL's and NTPC's Strong Positions:** Both BPCL and NTPC show high Interest Coverage Ratios, indicating strong financial positions and a robust ability to cover interest obligations.
- IOCL's and ONGC's Moderate Positions:** IOCL and ONGC have lower Interest Coverage Ratios, suggesting a comparatively moderate ability to cover interest expenses. Further analysis is needed to understand the factors influencing their financial positions.



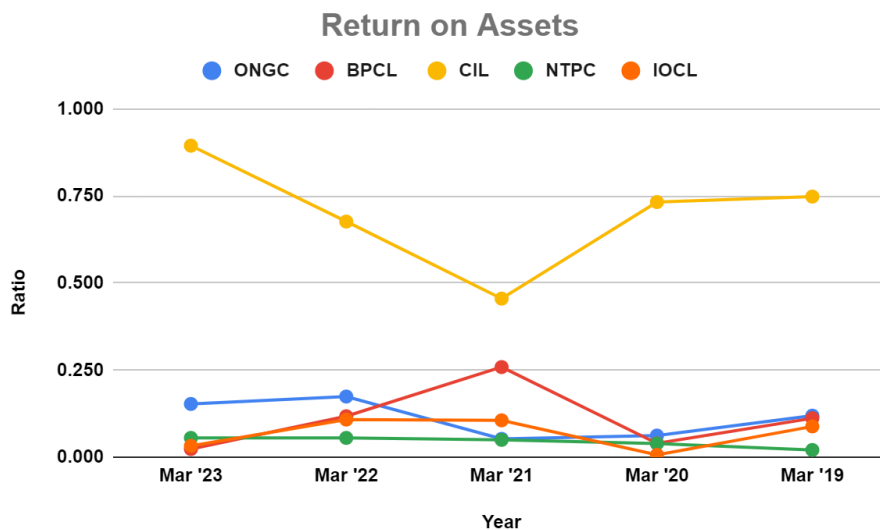
Inferences:

- **CIL's Exceptional Profitability:** CIL stands out with an extremely high profit margin, showcasing exceptional efficiency in generating profits from its revenue.
- **BPCL and NTPC's Solid Performance:** BPCL and NTPC demonstrate strong profit margins, suggesting efficient cost management and solid profitability.
- **IOCL's Moderate Profitability:** IOCL shows a moderate profit margin, indicating a positive but relatively lower efficiency in converting sales into profit. Further analysis can help identify areas for improvement.
- **From the data we observed that the Profit Margin has increased for some of the companies and decreased for others over the year with BPCL boasting the highest increase.**



Inferences:

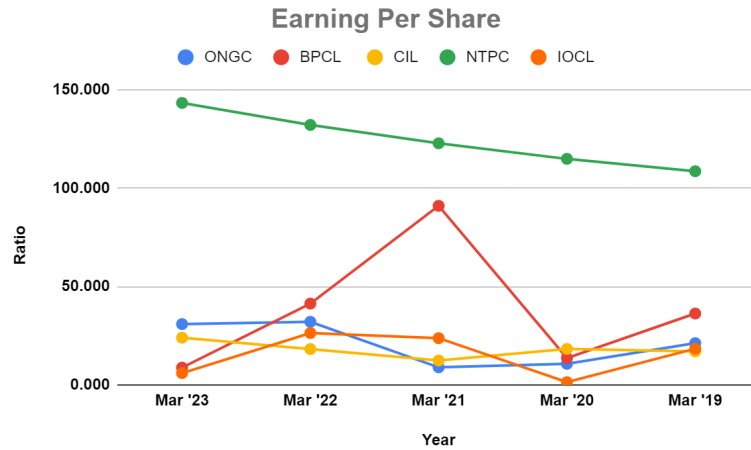
- Efficiency in Equity Utilisation: CIL and BPCL demonstrate efficient utilisation of shareholders' equity, resulting in high ROE values.
- Moderate Performance by IOCL: IOCL exhibits a moderate ROE, suggesting a reasonable ability to generate profit in relation to shareholders' equity.
- ONGC and NTPC's Lower ROE: ONGC and NTPC have lower ROE values, indicating potential areas for improvement in optimising profitability relative to shareholders' equity.
- From the data we observed that the Return on the Equity has increased for some of the companies and decreased for others over the year with CIL boasting the highest increase from 0.748 to 0.895.



Inferences:

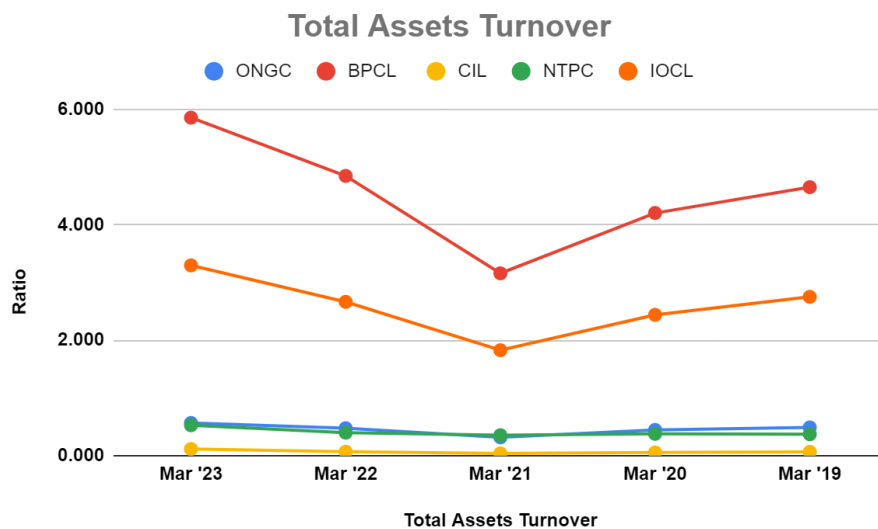
- Efficiency Variation: The significant difference in ROA values among the companies indicates varying degrees of efficiency in asset utilisation.
- CIL's Strong Performance: CIL stands out as a company with a remarkably high ROA, suggesting efficient management of its assets to generate profit.
- Areas for Improvement: BPCL and IOCL show lower ROAs, indicating potential areas for improvement in optimising asset utilisation and enhancing profitability.

- From the data we observed that the Return on the asset has increased for some of the companies and decreased for others over the year with CIL boasting the highest increase from 0.748 to 0.894.



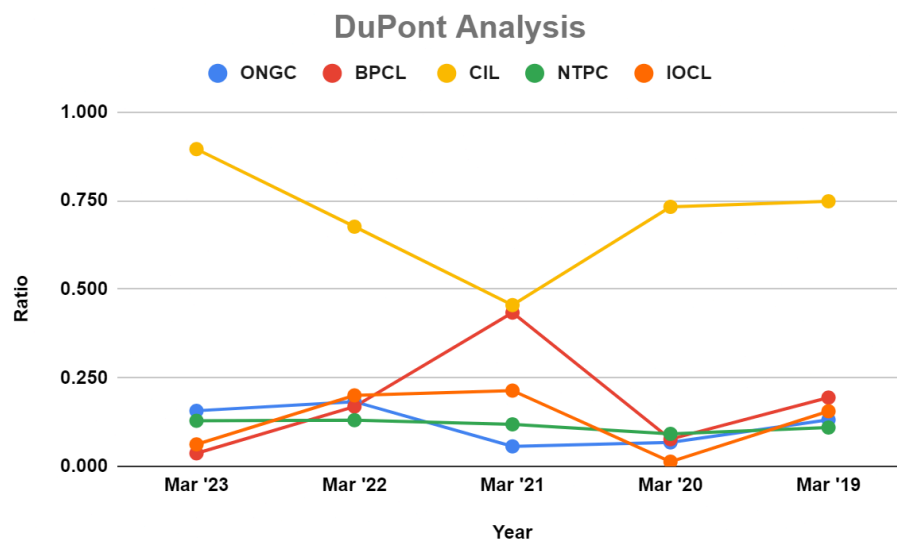
Inferences:

- NTPC's Strong Earnings:** NTPC demonstrates robust earnings on a per-share basis, reflecting strong profitability.
- ONGC and CIL's Healthy Performance:** ONGC and CIL also show healthy EPS values, indicating good profitability and earnings distribution to shareholders.
- BPCL and IOCL's Lower EPS:** BPCL and IOCL have lower EPS values, suggesting a need for further analysis to understand the factors impacting their profitability on a per-share basis.



Inferences:

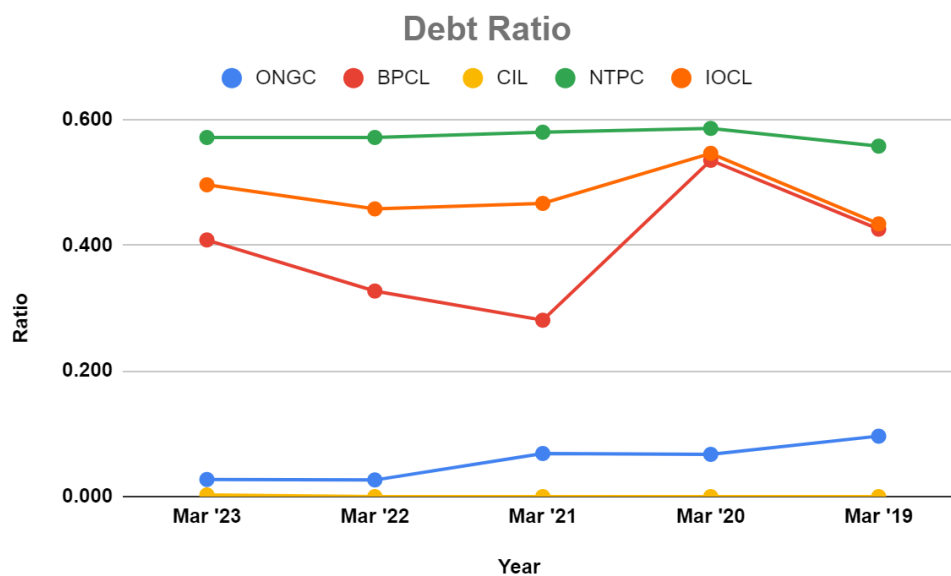
- BPCL and IOCL's Exceptional Efficiency: BPCL and IOCL stand out with high total asset turnovers, indicating superior efficiency in utilising assets to generate revenue.
- NTPC's Moderate Efficiency: NTPC shows a moderate total asset turnover, suggesting reasonable efficiency in asset utilisation for revenue generation.
- ONGC's and CIL's Comparable Efficiency: ONGC and CIL have comparable but moderate total asset turnovers, signifying a similar level of efficiency in converting assets into revenue.
- From the data we observed that the Total Asset Turnover has increased for some of the companies and decreased for others over the year with BPCL boasting the highest increase.



Inferences:

- CIL's Strong Performance: CIL stands out with a high ROE, indicating a strong overall performance in profitability, asset utilisation, and equity management.
- ONGC's Moderate Performance: ONGC demonstrates a moderate ROE, suggesting a balanced performance in profitability, asset utilisation, and equity management.

- **NTPC's Reasonable Performance:** NTPC shows a reasonable ROE, reflecting a balanced approach in profitability, asset utilisation, and equity management.
- **IOCL's and BPCL's Challenges:** IOCL and BPCL exhibit lower ROEs, indicating potential challenges in one or more components of the DuPont Analysis. Further analysis is needed to identify specific areas for improvement.
- **Holistic Financial Analysis:** DuPont Analysis provides a comprehensive view of the factors contributing to ROE, helping professionals identify areas for improvement and make informed financial decisions.



Inferences:

- **CIL and ONGC's Conservative Approach:** CIL and ONGC exhibit a conservative approach with very low debt ratios, implying lower financial risk and a preference for equity financing.
- **BPCL, IOCL, and NTPC's Reliance on Debt:** BPCL, IOCL, and NTPC have higher debt ratios, suggesting a greater reliance on debt for financing. While this can offer tax advantages, it also increases financial risk.
- **Financial Risk Considerations:** The debt ratios provide insights into the financial risk associated with each company. Companies with higher debt ratios may face greater challenges in managing debt-related obligations and interest payments.

- From the data we observed that the Debt Ratio has increased for some of the companies and decreased for others over the year with IOCL boasting the highest increase.

Question 3

Ranking Based on Financial Stability: Considering the given financial ratios, a comprehensive ranking in descending order of financial stability can be established:

1. Coal India Limited (CIL): Demonstrates consistently high performance across various ratios, indicating strong profitability, asset utilisation, and equity management.
2. Oil and Natural Gas Corporation (ONGC): Shows a mix of moderate and lower ratios but maintains a relatively stable financial position.
3. National Thermal Power Corporation (NTPC): Exhibits reasonable financial stability with a balanced performance across different metrics.
4. Indian Oil Corporation Limited (IOCL): Faces challenges in some ratios, indicating potential areas for improvement, but maintains moderate financial stability.
5. Bharat Petroleum Corporation Limited (BPCL): Shows lower rankings in several ratios, suggesting potential financial challenges and areas for improvement.